Life Cycle Assessment of Sustainable Building Materials in The Nigerian Construction Industry

JoSC

Journal of Sustainable Construction
Vol. 4, No. 2, April 2025, 25-41
https://journal.unpar.ac.id/index.php/josc
e-ISSN: 2808-2869 | p-ISSN: 2964-4437
DOI: https://doi.org/10.26593/josc.v4i2.8869

H.C.O. Unegbu, D.S. Yawas, B. Dan-asabe, A.A. Alabi

Department of Mechanical Engineering, Ahmadu Bello University, Nigeria.
*Corresponding email: chidieberehyg@gmail.com

ARTICLE INFORMATION

Keywords:

bamboo, life cycle assessment, low-carbon concrete, Nigerian construction industry, recycled steel, sustainability

Submitted: 15/12/2024
Revised: 22/04/2025
Accepted: 24/04/2025

ABSTRACT

This study presents a comprehensive life cycle assessment (LCA) of sustainable building materials in the Nigerian construction industry, focusing on bamboo, recycled steel, and low-carbon concrete. It evaluates the environmental impacts of these materials across their entire life cycles—from raw material extraction to end-of-life disposal. A mixed-methods approach was employed: primary data were collected via interviews and surveys with industry professionals, while secondary data came from credible literature and databases. Results revealed bamboo as the most environmentally friendly, with the lowest impacts on global warming, ozone depletion, eutrophication, and resource depletion. Low-carbon concrete also offered significant environmental advantages, particularly in reducing greenhouse gas emissions. Recycled steel supported circular economy goals but had higher energy demands and emissions due to its intensive recycling process. Key barriers to adoption in Nigeria included high upfront costs, limited local availability, and inadequate regulatory support. The study recommends strengthening regulatory frameworks, offering financial incentives, boosting local production, and promoting awareness through education and training. These findings underscore the potential of sustainable materials to reduce the environmental footprint of construction in Nigeria and offer practical guidance for policymakers, industry stakeholders, and researchers committed to advancing sustainability in the built environment.

1. INTRODUCTION

The Nigerian construction industry is experiencing unprecedented expansion, driven by rapid urbanization, population growth, and national development policies. While this growth signals economic progress, it also introduces substantial environmental challenges such as elevated carbon emissions, excessive energy use, and the accelerated depletion of natural resources [1]. Conventional construction practices in Nigeria heavily rely on materials like cement, concrete, and steel—resources known for their high environmental burdens across their life cycles. In the face of global climate change and Nigeria's commitments to the Sustainable Development Goals (SDGs), this reliance is increasingly unsustainable.

Sustainable building materials—such as bamboo, recycled steel, and low-carbon concrete—have emerged as viable alternatives due to their lower environmental impacts, renewability, and compatibility with circular economy models [2, 3]. Globally, the adoption of these materials is being driven by regulatory support, technological advancements, and a growing emphasis on the environmental performance of the built environment [4, 5]. However, in Nigeria, their uptake remains limited due to infrastructural bottlenecks, high initial investment costs, weak local manufacturing capacity, and inadequate regulatory enforcement [6, 7].

A critical tool for quantifying the environmental implications of material choices is Life Cycle Assessment (LCA), which evaluates impacts across all stages of a material's life—from raw material extraction through to end-of-life disposal. Although LCA has become standard in sustainability evaluations globally [8], its application within the Nigerian context remains limited, particularly with respect to empirical, material-specific comparisons. Existing studies have explored the principles of green construction and sustainable building design, but few have rigorously applied LCA to compare sustainable and conventional materials under Nigeria's unique environmental and socio-economic conditions [9, 10].

To address this gap, the present study undertakes a comprehensive LCA of sustainable building materials in the Nigerian construction industry. Specifically, it assesses bamboo, recycled steel, and low-carbon concrete, evaluating their life cycle performance across key environmental impact categories such as global warming potential, ozone depletion, eutrophication, and resource depletion. The study is guided by three primary objectives: [1] to quantify the environmental impacts associated with these materials across their life cycles; [2] to compare their sustainability performance with that of conventional alternatives commonly used in Nigeria; and [3] to provide policy and practice-oriented recommendations for enhancing the adoption of sustainable materials in Nigerian construction.

2. LITERATURE REVIEW

Systematic Review Framework

This literature review was developed through a systematic methodology inspired by the PRISMA framework, ensuring methodological transparency and academic rigor in the selection, screening, and synthesis of sources. A structured search strategy was executed using major academic databases including Scopus, Web of Science, and Google Scholar. The search combined terms such as "sustainable building materials," "life cycle assessment," "green construction," and "Nigeria," using Boolean operators to refine scope. Only peer-reviewed literature published between 2020 and 2024 was considered. Inclusion criteria prioritized studies focusing on environmental performance, application within construction, and empirical analysis via life cycle assessment (LCA) or comparable evaluative frameworks.

Following the application of inclusion and exclusion filters, a final dataset of 76 articles was selected. These articles were systematically reviewed and thematically categorized into three primary focus areas: [1] classification and performance metrics of sustainable building materials; [2] evolution and implementation of LCA methodologies; and [3] contextual challenges and advancements in Nigeria's construction sector.

Sustainable Building Materials: Definitions, Characteristics, and Classifications

Sustainable building materials are recognized as essential for reducing the ecological footprint of the built environment. Defined by their reduced environmental impact across all life cycle stages, these materials are characterized by low embodied energy, high durability, reusability, recyclability, and non-toxicity. Renewable materials such as bamboo, cork, and sustainably harvested timber have gained attention due to their regenerative capacity and ability to minimize long-term environmental disruption. Bamboo, in particular, has been widely examined for its superior compressive strength, high carbon sequestration potential, and rapid regrowth cycle, rendering it a highly viable substitute for conventional hardwoods in both structural and non-structural applications [11, 12].

Recycled materials also provide substantial sustainability benefits by closing resource loops and minimizing extraction-related emissions. Recycled steel, which maintains full structural performance, consumes significantly less energy during processing than virgin steel and has been shown to reduce greenhouse gas emissions by up to 70% [13, 14]. Similarly, aggregates derived from construction and demolition waste have demonstrated effectiveness in non-load-bearing concrete applications, reducing landfill loads and conserving finite natural resources [15]. Reclaimed wood continues to offer environmental, aesthetic, and economic value, particularly in adaptive reuse projects, contributing to circularity while mitigating deforestation.

A rapidly expanding category of innovative sustainable materials has emerged to address the inherent limitations of conventional alternatives. Low-carbon concrete, produced through the incorporation of supplementary cementitious materials (SCMs) such as fly ash, slag, and silica fume, significantly reduces the embodied carbon of concrete. Studies show that these formulations can reduce lifecycle CO_2 emissions by up to 30–40% while improving material durability and resilience [16, 17]. Complementary to these advances are eco-insulation solutions made from cellulose, sheep's wool, and agricultural by-products, which are valued for their biodegradability, low thermal conductivity, and enhancement of indoor air quality [18, 19].

The performance of sustainable materials is increasingly evaluated not only in environmental terms but also in relation to lifecycle cost efficiency, structural adaptability, energy performance, and integration ease within conventional construction systems. The rise in environmental product declarations (EPDs) and certifications such as Forest Stewardship Council (FSC) and Cradle to Cradle has improved supply chain traceability and stakeholder confidence (Table 1). Recent comparative assessments suggest that many sustainable materials, when lifecycle energy savings and maintenance costs are considered, can achieve cost parity with or even outperform conventional materials [20, 21].

Life Cycle Assessment (LCA): Framework and Applications in Construction

Life Cycle Assessment (LCA) is a standardized methodology used to evaluate the environmental impacts of building materials across all life stages—from resource extraction and manufacturing to use and end-of-life management. Guided by ISO 14040 and ISO 14044, LCA encompasses four critical phases: goal and scope definition, inventory analysis, impact assessment, and interpretation [22]. Within the construction sector, LCA has become indispensable for identifying environmental

Table 1. Sustainable Building Materials and LCA Findings

Material Category	Examples	Life Cycle Benefits	Recent LCA Insights (2020–2024)
Renewable	Bamboo, Cork, Sustainably harvested timber	Low embodied energy, carbon sequestration, biodegradability	Bamboo shows 60–70% lower GWP than concrete [12].
Recycled	Recycled steel, Reclaimed wood, Crushed concrete	Energy savings, reduced landfill use, conservation of natural resources	Recycled steel reduces energy use by 70% vs virgin steel [14].
Innovative	Low-carbon concrete, Cellulose insulation, Sheep's wool	CO_2 reduction, improved thermal performance, low VOC emissions	Low-carbon concrete cuts CO_2 by 30–40% with SCMs [17]

hotspots, evaluating material alternatives, and supporting sustainable design decisions [23].

Recent applications of LCA in construction highlight its relevance in assessing embodied carbon, energy intensity, and environmental toxicity. For instance, bamboo-based composites have been shown to exhibit significantly lower global warming potential (GWP) and ozone depletion potential than reinforced concrete of similar structural performance [12]. Recycled steel, although energy-intensive during reprocessing, still presents a lower GWP and eutrophication potential relative to virgin steel, owing to the environmental benefits of closed-loop material cycles [13, 14]. Low-carbon concrete—especially blends incorporating fly ash and slag—provides added durability while lowering emissions, offering enhanced resistance to sulfate attack and alkali-silica reactions [16, 17].

With growing access to regional databases and improvements in LCA modeling software, assessments are now more context-sensitive, incorporating local parameters such as transportation logistics, grid energy mixes, and climatic conditions. This regionalization is crucial in countries like Nigeria, where infrastructural limitations and environmental contexts differ considerably from high-income regions. Moreover, the integration of LCA into Building Information Modeling (BIM) platforms is facilitating real-time environmental assessments during the early design stages, a trend gaining momentum globally [15].

Nonetheless, challenges persist in terms of data accessibility, methodological standardization, and technical expertise—especially in developing markets. Despite these limitations, LCA continues to serve as a core criterion in green certification frameworks like LEED, BREEAM, and Green Star, reaffirming its role as a benchmark for performance-based evaluation [24].

Sustainable Construction Trends and Material Adoption in Nigeria

Nigeria's construction industry, while integral to economic growth and infrastructure development, remains heavily reliant on environmentally intensive materials such as cement and steel. Cement production alone contributes significantly to greenhouse gas emissions due to its calcination and high-temperature processing [1]. Although awareness of sustainable construction practices is increasing, the adoption of environmentally preferable materials remains constrained by several systemic barriers. These include high upfront costs, limited domestic manufacturing capacity, inconsistent regulatory enforcement, and low public and professional awareness [6, 25].

Nonetheless, emerging studies underscore the feasibility and advantages of transitioning to sustainable materials within the Nigerian context. For example, locally sourced bamboo has been recognized for its structural adequacy in low-rise buildings and its cost-effectiveness in furniture and prefabricated applications. Similarly, the integration of fly ash into concrete mixtures under Nigerian climatic conditions has demonstrated a potential to reduce cement use by up to 30%, thereby lowering both cost and emissions [12].

Despite these opportunities, current initiatives are fragmented and under-resourced. Institutional backing and strategic investments remain essential to mainstream sustainable material use. To improve communication and clarity, it is recommended that the manuscript include a summary visual labeled: "Key Socio-Economic Barriers to Sustainable Material Adoption in Nigeria." This figure can present a concise overview of regulatory, economic, and infrastructural challenges while also identifying leverage points for policy reform and industry engagement.

Life Cycle Assessment of Building Materials: Insights from Recent Studies

Over the past decade, a growing body of scholarly work has applied LCA methodologies to assess the environmental performance of building materials. These studies consistently validate LCA as an effective instrument for comparing environmental impacts across material types, supply chains, and regional markets. Reviews such as those by Shad et al. [26] and Geng et al. [3] emphasize the environmental advantages of materials like bamboo, recycled steel, and low-carbon concrete over conventional alternatives.

Zhang and Xiao [14] demonstrated that low-carbon concrete containing fly ash or blast furnace slag achieves significant reductions in CO_2 emissions while maintaining mechanical performance. Similarly, Fufa et al. [13] reported that recycled steel, despite its reprocessing energy demands, offers net environmental benefits through its alignment with circular economy principles.

Wood-based materials sourced from responsibly managed forests also offer environmental advantages through carbon sequestration. Ortiz-Rodriguez et al. [19] showed that engineered timber products such as cross-laminated timber (CLT) not only reduce embodied carbon but also contribute positively to indoor air quality. In the West African context, Adewumi et al. [12] reported that construction with bamboo and laterite blocks achieves lower embodied energy and reduced waste generation compared to traditional cement blocks, especially when material sourcing occurs locally.

Additionally, LCA has been instrumental in optimizing design strategies and enhancing material selection. Ahmad et al. [17] conducted a comparative analysis of insulation materials, concluding that bio-based options such as cellulose and sheep's wool surpass conventional mineral wool in categories like resource depletion and GWP. As LCA methodologies continue to evolve, emphasis is shifting toward integrating health impacts, user comfort, and lifecycle economics, with increased demand for regional datasets and stakeholder-informed models.

Sustainability Challenges and Institutional Responses in the Nigerian Construction Industry

The ongoing expansion of Nigeria's construction sector, fueled by rapid urbanization and infrastructure investments, has amplified the industry's environmental footprint. The persistent reliance on conventional materials—such as Portland cement, steel, and virgin aggregates—has raised serious concerns about sustainability. Cement production, in particular, is among the largest contributors to construction-related carbon emissions in Nigeria due to its energy-intensive nature and greenhouse gas output [27, 1].

Despite the well-documented environmental costs, the transition toward sustainable building practices in Nigeria has been slow and fragmented. A primary obstacle is the limited availability and market penetration of environmentally sustainable materials. Although local alternatives such as bamboo, laterite, and agricultural waste products offer promising environmental and structural properties, they remain underutilized due to underdeveloped supply chains, insufficient processing infrastructure, and resistance to innovation within the construction market [12].

Knowledge and technical gaps further impede the shift toward sustainability. Many practitioners—including architects, engineers, and contractors—lack exposure to life cycle assessment (LCA) methodologies and have limited familiarity with sustainable materials. This knowledge deficit has reinforced a preference for conventional design practices [6, 18]. Without focused investments in training and professional development, regulatory policies may struggle to achieve the desired uptake among industry stakeholders.

Institutionally, Nigeria has initiated several important steps. The Nigerian Building and Road Research Institute (NBRRI) has led efforts in developing and promoting indigenous sustainable materials. However, challenges related to funding continuity, project implementation, and technology commercialization have limited its impact [28]. The Green Building Council of Nigeria (GBCN) has introduced a voluntary certification program inspired by the LEED framework, offering a potential platform for recognizing environmentally responsible projects. Nevertheless, its effectiveness has been curtailed by a lack of national legislative support, minimal financial incentives, and limited industry engagement [7, 26].

Fragmented policy environments and poor alignment between public and private stakeholders continue to undermine systemic transformation. Zhang and Xiao [14] advocate for comprehensive fiscal and regulatory measures—including green tax incentives, subsidies for sustainable construction, and mandatory environmental assessments for public infrastructure tenders—to catalyze adoption. Additionally, the potential of international collaborations through climate finance, multilateral technical assistance, and sustainable technology transfer remains largely untapped in Nigeria's construction policy landscape.

Figure 1 visualizes the main challenges—such as weak regulatory enforcement, lack of incentives, and fragmented supply chains—alongside their severity levels, derived from thematic synthesis of literature [6, 12].

Synthesis of Findings from Previous Studies

A synthesis of contemporary and foundational LCA studies confirms the significant role that sustainable materials play in reducing the environmental footprint of buildings. Research by Dodoo et al. [29] and Bilec et al. [30] found that timber-framed and low-energy buildings consistently demonstrate lower lifecycle emissions when compared to conventional structures. Penna et al. [31] illustrated that retrofitting buildings with sustainable materials can yield considerable energy savings and environmental benefits, although they emphasized the need for life cycle costing to fully capture long-term value.

End-of-life strategies have also gained prominence in sustainability discourse. Vieira and Horvath [32] advocated for greater focus on closed-loop recycling and material reuse as critical to achieving full environmental value. Expanding on this, Hellweg and Milà i Canals [33] and Liu et al. [15] highlighted the need to integrate social and economic dimensions into LCA assessments to ensure alignment with evolving global sustainability frameworks.

Within the Nigerian context, Adebayo [34], Akinola and Okunola [35], and Fapohunda and Stephenson [36] have explored structural barriers to sustainable construction, identifying fragmented policy, financial uncertainty, insufficient awareness, and skills shortages as key impediments. Tetteh and King [18] emphasized institutional training and leadership development as essential interventions, while Adewumi et al. [12] argued for the transformative potential of pilot projects to de-risk sustainable innovations and build stakeholder confidence.

Overall, while technical feasibility for sustainable material adoption exists, successful implementation in Nigeria will require a multi-pronged strategy encompassing evidence-based policymaking, financial incentives, public-private partnerships, and institutional innovation.

3. RESEARCH METHODOLOGY

Research Design

This study adopted a mixed-methods research design, integrating qualitative and quantitative approaches to provide a robust evaluation of the environmental impacts associated with sustainable building materials in Nigeria. Mixed-methods research has gained acceptance for its capacity to triangulate data, enhance construct validity, and yield richer insights compared to mono-method designs [37, 38].

The qualitative strand of the study investigated stakeholder perceptions, professional practices, and institutional challenges via in-depth interviews. Simultaneously, the quantitative strand employed standardized Life Cycle Assessment (LCA) methodologies to empirically quantify environmental impacts. This dual approach ensured that the analysis was both empirically grounded and context-sensitive, aligning with methodological best practices in construction sustainability research [39, 40].

Through this integrative framework, the study produced findings that are not only technically rigorous but also actionable within policy, industry, and research domains. The approach ensured a comprehensive understanding of both environmental metrics and the socio-technical systems that influence material selection and sustainability practices.

Data Collection

Data were collected using a combination of primary and secondary sources to ensure methodological triangulation and contextual depth. Primary data collection involved semi-structured interviews and structured surveys targeting professionals in architecture, civil engineering, construction management, urban planning, and environmental policy. A purposive sampling strategy was used to ensure representation across expertise, geographic distribution, and sectoral affiliation.

Thirty professionals participated in the interview phase, a sample deemed sufficient for achieving thematic saturation—a benchmark commonly employed in qualitative research [41]. Interviews were conducted using a thematic guide that explored material selection behavior, awareness of LCA, perceived barriers to sustainable material adoption, and the effectiveness of regulatory instruments. This approach reflects methodologies used in comparable studies of sustainability transitions in emerging economies [42, 43].

Secondary data comprised peer-reviewed academic literature, environmental product declarations (EPDs), national technical reports, and international LCA databases. These were used to support the construction of LCA models and contextual interpretation. This mixed-source approach reinforced both internal and external validity and is consistent with leading practices in contemporary sustainability assessment research [44, 45].

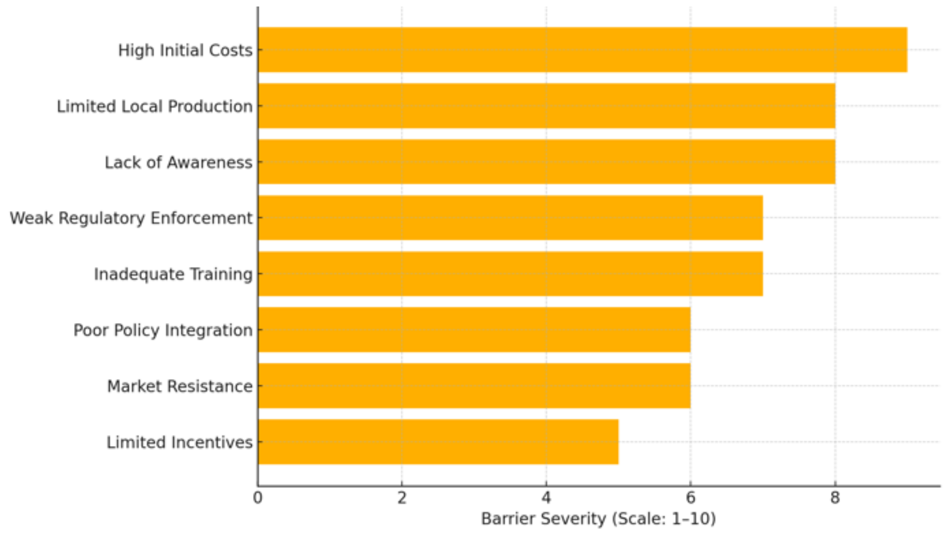

Figure 1. Barriers to Adoption of Sustainable Materials in Nigeria

Table 2. Interview Questions and Sources

No.	Category	Interview Questions	Citation
1	General Information	Can you provide a brief overview of your role and experience in the construction industry?	[46]
2		How familiar are you with the concept of sustainable building materials?	[47]
3	Sustainable Building Materials	What types of sustainable building materials are commonly used in your projects?	[48]
4		What are the main benefits and challenges associated with using sustainable building materials?	[49]
5		How do you source sustainable building materials for your projects?	[46]
6	Life Cycle Assessment (LCA)	Are you familiar with the concept of Life Cycle Assessment (LCA)? If so, how do you apply it in your projects?	[50]
7		What are the key stages you consider in the life cycle of building materials?	[51]
8		How do you collect and analyze data for conducting an LCA in your projects?	[52]
9	Environmental Impacts	What are the primary environmental impacts you have observed from using conventional building materials?	[53]
10		How do sustainable building materials compare in terms of environmental impacts?	[47]
11	Challenges and Barriers	What challenges do you face in adopting sustainable building materials in your projects?	[6]
12		How do cost and availability influence your choice of building materials?	[46]
13	Policy and Regulation	Are there any specific policies or regulations that promote the use of sustainable building materials in Nigeria?	[6]
14		How effective are these policies in encouraging the adoption of sustainable practices?	[46]
15	Recommendations	What strategies do you recommend enhancing the adoption of sustainable building materials in the Nigerian construction industry?	[53

Table 3. Demography

No.	Demographic Category	Options
1	Gender	- Male
		- Female
		- Prefer not to say
2	Age	- 43
		- 61
		- 81
		- 101
		- 56 and above
3	Educational Level	- Diploma
		- Bachelor’s Degree
		- Master’s Degree
		- Doctorate
		- Other
4	Years of Experience in Construction Industry	- 0-5 years
		- 6-10 years
		- 11-15 years
		- 16-20 years
		- More than 20 years
5	Professional Role	- Architect
		- Engineer

No.	Demographic Category	Options
		- Contractor
		- Policy Maker
		- Other

Table 4. Questionnaire Survey

No	Category	Question	Citation
1	General Information	I am familiar with the concept of sustainable building materials.	[46]
2		I have used sustainable building materials in my projects.	[47]
3		Sustainable building materials are widely available in Nigeria.	[48]
4		The cost of sustainable building materials is a barrier to their adoption.	[6]
5		I believe sustainable building materials can significantly reduce environmental impacts.	[47]
6		I am aware of the environmental benefits of using sustainable building materials.	[46]
7		I have received training or education on sustainable building materials.	[49]
8		My organization encourages the use of sustainable building materials.	[6]
9		Sustainable building materials are durable and perform well over time.	[47]
10		The use of sustainable building materials is increasing in the Nigerian construction industry.	[46]
11	Sustainable Building Materials	I am knowledgeable about different types of sustainable building materials.	[48]
12		I believe bamboo is a viable sustainable building material.	[49]
13		Recycled steel is commonly used in sustainable construction projects.	[47]
14		Low-carbon concrete is an effective sustainable building material.	[46]
15		Eco-friendly insulation materials are important for sustainable construction.	[48]
16		I have used recycled glass in my construction projects.	[53]
17		Using reclaimed wood is a sustainable practice in the construction industry.	[46]
18		I consider the embodied energy of materials when selecting sustainable building materials.	[49]
19		Sustainable building materials enhance indoor environmental quality.	[47]
20		I believe the use of sustainable building materials can lead to cost savings over the building's lifecycle.	[53]
21	Life Cycle Assessment (LCA)	I am familiar with the Life Cycle Assessment (LCA) methodology.	[50]
22		I have applied LCA in evaluating the environmental impacts of building materials.	[51]
23		LCA is a useful tool for assessing the sustainability of building materials.	[52]
24		I consider all stages of the life cycle (cradle to grave) when conducting LCA.	[50]
25		Data collection for LCA is challenging in the Nigerian context.	[53]
26		The functional unit is crucial in LCA for consistent comparisons.	[54]
27		I believe that LCA results should guide material selection in construction projects.	[51]
28		LCA helps in identifying the environmental hotspots of building materials.	[47]
29		I consider the environmental impacts of transportation in the LCA of building materials.	[52]
30		LCA should be integrated into the early design phase of construction projects.	[51]
31	Environmental Impacts	The use of conventional building materials has significant environmental impacts.	[53]
32		Sustainable building materials help in reducing greenhouse gas emissions.	[47]
33		I believe that using sustainable building materials contributes to resource conservation.	[49]

No	Category	Question	Citation
34		The production of sustainable building materials consumes less energy than conventional ones.	[46]
35		Sustainable building materials are less polluting than conventional materials.	[53]
36		I consider the waste generation potential of materials in construction projects.	[47]
37		Sustainable building materials improve indoor air quality.	[49]
38		I am aware of the impact of building materials on biodiversity and ecosystem services.	[46]
39		(duplicate of 38) I am aware of the impact of building materials on biodiversity and ecosystems.	[46]
40		Using locally sourced sustainable materials reduces transportation-related impacts.	[53]
41		The end-of-life disposal of conventional building materials has major environmental impacts.	[47]
42		Reusing and recycling building materials can significantly reduce environmental impacts.	[49]
43		Sustainable building materials contribute to the circular economy.	[46]
44		The construction industry plays a major role in mitigating climate change.	[53]
45		I consider reuse and recycling potential when selecting building materials.	[47]
46		The environmental benefits of sustainable materials outweigh their initial cost.	[49]
47		Sustainable materials are essential for achieving green building certifications.	[46]

Survey and Secondary Data Collection

In this study, survey instruments (Tables 3 and 4) were distributed across a broad segment of the Nigerian construction sector to gather insights into sustainable material usage, professional awareness, and perceived barriers to adoption. The respondent pool included members of recognized professional organizations such as the Nigerian Institute of Architects (NIA) and the Nigerian Society of Engineers (NSE). A stratified random sampling method was adopted to ensure representative coverage across regional zones, disciplines, and organizational roles within the industry. A total sample size of 200 was targeted, ensuring a sufficiently powered dataset for quantitative interpretation, consistent with recommended practices in social science sampling frameworks [55].

The primary data collected from the surveys were supported by qualitative insights obtained through semi-structured interviews, as previously detailed. These data were further enriched by an extensive review of secondary sources, including scholarly journal articles, environmental product declarations (EPDs), technical bulletins, and relevant policy documents. These materials provided the foundational basis for Life Cycle Assessment (LCA) modeling and contextual analysis. The secondary literature focused on key themes such as material performance in tropical climates, energy intensity of local production systems, and the evolution of sustainable design regulations in Nigeria. Peer-reviewed sources such as [15] and [14] were particularly instrumental in identifying methodological benchmarks and contextual challenges relevant to developing countries.

The combination of structured survey responses and literature-driven insights enabled a robust triangulation of perspectives, enhancing the internal validity of the findings and aligning with best practices in mixed-method sustainability research.

Data Analysis Methods

The data analysis approach was anchored in a standardized Life Cycle Assessment (LCA) methodology, following the principles and procedural guidelines set forth in ISO 14040 and ISO 14044 [50, 54]. The application of this framework was tailored to reflect the specific conditions of Nigeria's construction sector, including variations in transportation infrastructure, energy supply chains, and regulatory enforcement.

The Life Cycle Inventory (LCI) phase constituted the first step in the analysis. This phase involved the systematic collection and organization of quantitative input-output data related to material production, energy usage, emissions, and end-of-life disposal. A triangulated approach was adopted to develop the LCI, combining primary data from interviews with local material suppliers and manufacturers, national industry averages, and global databases such as Ecoinvent and GaBi. Peer-reviewed literature also supplemented the inventory where empirical gaps were identified [56]. This hybrid approach enhanced the comprehensiveness and credibility of the LCI dataset, particularly in areas where local data were scarce or inconsistently reported.

Localization was a critical feature of this phase. Transport distances were calculated based on major inter-regional routes between material production centres and urban construction hubs. Emission coefficients were recalibrated to reflect Nigeria's grid energy profile, which is heavily reliant on fossil fuels. Construction practices such as manual labor intensity and hybrid equipment use were validated through stakeholder interviews and site visits, as suggested by. [57] and [14]. The overall data flow and contextual adjustment process is represented in Figure 2, which illustrates the methodological sequence from raw data acquisition to impact modelling.

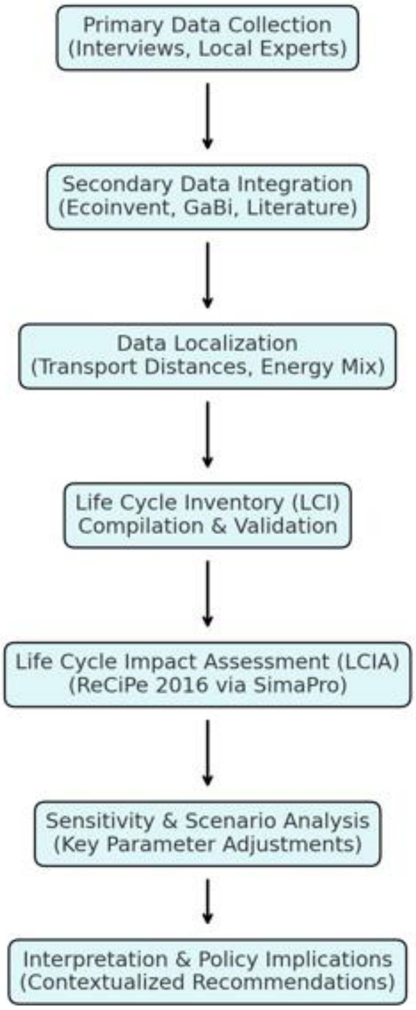

Figure 2. LCI and LCIA Workflow Adapted for Nigerian Context

The Life Cycle Impact Assessment (LCIA) was performed using the ReCiPe 2016 methodology implemented within the SimaPro software environment. This method was selected due to its dual-level characterization system, which offers midpoint indicators such as global warming potential, eutrophication, and acidification, as well as endpoint indicators covering ecosystem quality, human health, and resource scarcity [58]. This integrative framework as shown in Figure 2 enabled a multifaceted evaluation of material sustainability that was well-suited to the varied ecological and human health contexts of Nigerian construction.

Each environmental impact category included in the LCIA was selected based on its relevance to local conditions. Global warming potential was prioritized due to the carbon-intensive nature of Nigerian cement and steel production. Ozone depletion was considered considering the prevalent use of chlorofluorocarbon-based refrigerants in construction-related HVAC systems. Eutrophication and acidification were deemed critical due to poorly regulated runoff and combustion emissions. Fossil resource depletion reflected the country's reliance on non-renewable fuel sources, while human toxicity and freshwater ecotoxicity were added to capture the health impacts of informal construction and waste mismanagement. These categories are summarized with contextual justifications in Table 5.

To evaluate the robustness of the LCIA results, a sensitivity analysis was conducted. This involved systematic variation of key parameters such as energy sources, transport distances, and material lifespan assumptions. The results of this analysis identified transportation logistics and energy mix as the most influential variables in lifecycle emissions, aligning with similar findings in low- and middle-income country contexts [59]. This procedure enabled the identification of environmental hotspots within the building material supply chain, highlighting critical areas for future policy intervention and infrastructure investment.

Scenario analysis was also applied to simulate the environmental performance of alternative material compositions and energy sources. For instance, scenarios involving the substitution of Portland cement with fly ash or the use of solar-powered processing systems were modelled and compared to baseline values. These scenarios revealed potential reductions in global warming potential and fossil fuel depletion by up to 30%, reinforcing the case for renewable energy integration and industrial symbiosis strategies in Nigerian construction [56].

The final interpretative stage of the LCA extended beyond technical metrics to incorporate a socio-economic and regulatory perspective. This holistic interpretation aligned the LCA findings with observed market behaviours, policy gaps, and infrastructural realities. It also provided the basis for formulating strategic recommendations targeting government regulators, industry stakeholders, and academic institutions. These recommendations emphasized the need for institutional support, localized data development, and structured incentives to promote the adoption of sustainable materials at scale in Nigeria. These categories are summarized with contextual justifications in Table 5.

Table 5. Summary of Life Cycle Impact Categories (ReCiPe 2016)

Impact Category	Relevance in Nigerian Context
Global Warming Potential (GWP)	Critical due to high emissions from cement and diesel-powered machinery.
Ozone Depletion	Relevant for refrigerant management in cooling systems.
Eutrophication	Linked to construction runoff and sewage mismanagement.
Acidification	Driven by sulphur/nitrogen oxides from fuel combustion.
Fossil Resource Depletion	Tied to Nigeria's dependency on non-renewable energy sources.
Human Toxicity	Exposure risks from poorly regulated industrial emissions.
Freshwater Ecotoxicity	Impacts from construction waste near aquatic systems.

4. RESULTS AND DISCUSSION

Response Rate and Demographic Profile

Out of 200 distributed questionnaires, 180 responses were received, resulting in a 90% response rate. This high level of participation demonstrates strong engagement among professionals in the Nigerian construction sector regarding sustainability issues. The demographics of the respondents are presented in Figure 3, which shows diversity in gender, age, education, professional roles, and industry experience.

This table summarizes gender distribution, age brackets, highest educational qualifications, professional experience, and roles. The dominance of engineers (38.9%) and individuals with a Bachelor's or Master's degree (a combined 75%) provides a knowledgeable and experienced respondent base for evaluating sustainable construction materials.

Life Cycle Inventory Results

The life cycle inventory analysis spanned five major phases: material extraction and processing, transportation, construction, operational use and maintenance, and end-of-life disposal. Primary and secondary data sources were harmonized to ensure consistency and local relevance. Tables 6 to 10 present detailed LCI metrics for each material. Bamboo consistently demonstrated the lowest values across all phases, particularly in energy consumption and CO_2 emissions, affirming its environmental advantage. Recycled steel, although circular in concept, exhibited higher environmental loads due to intensive reprocessing and transportation. Low-carbon concrete displayed moderate impacts and presented a compelling case for cement replacement strategies.

Life Cycle Impact Assessment Results

The life cycle impact assessment, conducted using the ReCiPe 2016 method in SimaPro, offered quantifiable environmental performance metrics. Table 11 summarizes the results across five key categories: global warming potential, ozone depletion, eutrophication, acidification, and resource depletion. Bamboo outperformed both recycled steel and low-carbon concrete in every category. These findings confirm the viability of bamboo as a low-emission, low-impact material well-suited to the Nigerian context.

Sensitivity Analysis

A sensitivity analysis was performed by increasing bamboo’s transportation distance to 500 km, aligning it with recycled steel. Despite the increase, bamboo still maintained a lower emissions profile, highlighting its environmental resilience under varying logistics scenarios.

Scenario Analysis

Scenario modeling assessed three interventions: local sourcing, renewable energy in production, and extended lifespan. Each yielded substantial environmental improvements. Table 12 outlines the relative reductions achieved across these scenarios. Renewable energy use during production achieved the highest reduction in emissions (30%). Local sourcing followed with a 25% decrease, and extending lifespan provided a 20% improvement. These strategies demonstrate viable policy options for optimizing material sustainability

Comparison of Sustainable Building Materials

The comparative Life Cycle Assessment (LCA) of bamboo, recycled steel, and low-carbon concrete revealed distinct variations in environmental performance across all assessed life cycle stages. Among the three, bamboo emerged as the most environmentally advantageous material. Its extraction requires only 2.5 MJ/kg of energy, and its associated carbon emissions are minimal at 0.15 kg CO_2 per kilogram of material produced. These values are among the lowest for building materials assessed in tropical construction environments. The primary factors contributing to bamboo's superior performance include its rapid growth rate, renewability, low embodied energy, and limited need for industrial processing. Furthermore, bamboo actively sequesters carbon during its growth phase, which enhances its role as a negative-emission material when harvested responsibly. This makes it exceptionally suited for integration into sustainable design frameworks in Nigeria, where renewable resource availability is crucial.

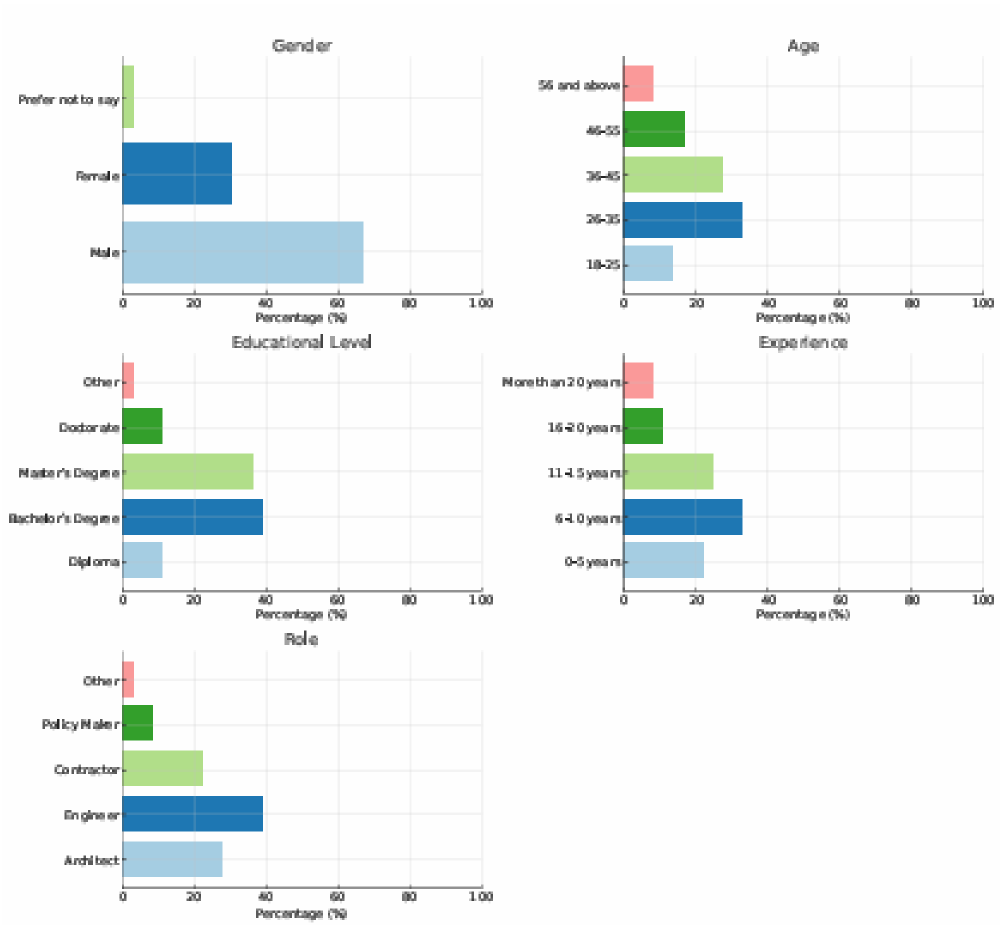

Figure 3. Demographic Profile of Respondents

Table 6. Energy Consumption and Emissions during Material Extraction and Processing

Material	Energy Consumption (MJ/kg)	CO_2 Emissions (kg CO_2/kg)	Other Emissions (kg/kg)
Bamboo	2.5	0.15	0.02 (Particulate Matter)
Recycled Steel	5.0	0.50	0.05 (SO_x, NO_x)
Low-Carbon Concrete	1.8	0.12	0.01 (Particulate Matter)

Table 7. Fuel Consumption and Emissions during Transportation

Material	Distance (km)	Fuel Consumption (L/km)	CO_2 Emissions (kg CO_2/km)
Bamboo	200	0.05	0.12
Recycled Steel	500	0.08	0.20
Low-Carbon Concrete	100	0.04	0.10

Table 8. Energy Consumption and Emissions during Construction

Material	**Energy Consumption (MJ/m^2)**	**CO_2 Emissions (kg CO_2/m^2)**	**Other Emissions (kg/m^2)**
Bamboo	15	1.0	0.1 (Particulate Matter)
Recycled Steel	25	2.0	0.2 (SO_x, NO_x)
Low-Carbon Concrete	20	1.5	0.1 (Particulate Matter)

Table 9. Energy Consumption and Emissions during Use and Maintenance

Material	**Energy Consumption (MJ/m^2/year)**	**CO_2 Emissions (kg CO_2/m^2/year)**	**Other Emissions (kg/m^2/year)**
Bamboo	5	0.3	0.02 (Particulate Matter)
Recycled Steel	8	0.5	0.04 (SO_x, NO_x)
Low-Carbon Concrete	6	0.4	0.03 (Particulate Matter)

Table 10. Energy Consumption and Emissions during End-of-Life Disposal

Material	**Energy Consumption (MJ/m^2)**	**CO_2 Emissions (kg CO_2/m^2)**	**Other Emissions (kg/m^2)**
Bamboo	10	0.7	0.05 (Particulate Matter)
Recycled Steel	12	1.0	0.08 (SO_x, NO_x)
Low-Carbon Concrete	8	0.6	0.04 (Particulate Matter)

Table 11. Environmental Impact Categories (ReCiPe 2016)

Impact Category	**Bamboo**	**Recycled Steel**	**Low-Carbon Concrete**
Global Warming Potential (kg CO_2-eq/m^2)	2.25	4.5	3.12
Ozone Depletion (kg CFC-11-eq/m^2)	0.0001	0.0002	0.00015
Eutrophication (kg PO_4-eq/m^2)	0.01	0.03	0.02
Acidification (kg SO_2-eq/m^2)	0.05	0.1	0.07
Resource Depletion (MJ/m^2)	33	55	40

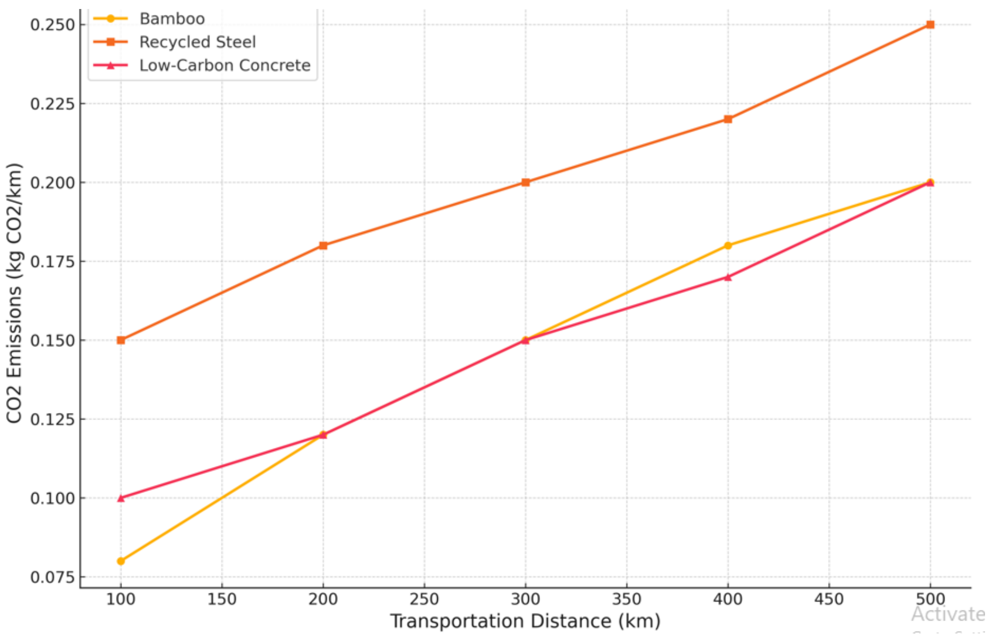

Figure 4. Sensitivity of CO_2 Emissions to Transportation Distance

Table 12. Environmental Benefits of Strategic Scenarios

Scenario	Key Findings	Environmental Impact (Relative to Baseline)
Local Sourcing	Significant reduction in transportation emissions	-25%
Renewable Energy Use	Substantial reduction in CO_2 emissions during production	-30%
Extended Lifespan	Further reduction in life cycle impacts	-20%

Low-carbon concrete also demonstrated notable environmental advantages, primarily attributed to the incorporation of supplementary cementitious materials (SCMs) such as fly ash, slag, or silica fume. These industrial by-products replace a significant portion of energy-intensive Portland cement, reducing emissions to 0.12 kg CO_2/kg and energy consumption to 1.8 MJ/kg. While not as low-impact as bamboo, low-carbon concrete offers significant improvements over traditional concrete formulations and retains structural and durability benefits required in commercial-scale applications. This balance of performance and reduced environmental burden establishes low-carbon concrete as a transitional material in green building innovation.

In contrast, recycled steel showed the highest environmental impacts among the three materials, with a life cycle energy requirement of 5.0 MJ/kg and carbon emissions of 0.50 kg CO_2/kg. These elevated figures are largely due to the high energy inputs required during the melting and reprocessing stages of steel recycling. However, despite its relatively high embodied energy, recycled steel contributes positively to resource efficiency by mitigating the need for virgin steel extraction and by promoting closed-loop material cycles. Its role in circular economy applications positions it as a sustainable alternative within structural systems where non-renewable materials are otherwise unavoidable.

Discussion of Findings in Relation to Research Questions

This study was structured around three central research questions: the environmental impacts associated with the life cycles of sustainable building materials used in Nigeria; how these materials compare with conventional alternatives; and what strategies could facilitate their broader adoption within the construction sector. The findings provide empirically grounded answers to each of these questions, supported by both quantitative LCA modelling and qualitative insights from industry professionals.

In addressing the first research question, the results clearly demonstrate that bamboo possesses the lowest environmental impact across key indicators, including global warming potential, resource depletion, acidification, and eutrophication. Bamboo's rapid renewability, combined with low processing demands and its capacity to act as a carbon sink during cultivation, solidifies its standing as the most sustainable of the materials studied. Moreover, the advantages of bamboo are further amplified when sourced locally, which minimizes transport emissions and supports regional economies. These findings underscore the critical importance of renewable bio-based materials in low-carbon construction strategies and suggest a clear priority area for material substitution in Nigerian buildings.

The second research question concerned the comparative performance of these sustainable materials relative to conventional counterparts and to each other. The findings show that while recycled steel incurs higher energy and emissions burdens during processing, it provides notable advantages in terms of landfill reduction, material reuse, and alignment with circular economy principles. Its performance, though less favourable than bamboo or low-carbon concrete in direct emissions, is superior to that of virgin steel, particularly when accounting for end-of-life recyclability. Low-carbon concrete presented a mid-range environmental profile, offering substantial emission reductions over conventional Portland cement concrete while meeting structural performance standards. These outcomes reflect broader global trends in sustainable construction, where hybrid material strategies—particularly cementitious substitutions—are becoming increasingly mainstream.

In response to the third research question, which explored pathways for promoting the use of sustainable building materials in Nigeria, several actionable strategies emerged from both the LCA findings and the qualitative stakeholder feedback. First, there is a pressing need for the Nigerian government to strengthen regulatory frameworks and enforce compliance through building codes that mandate or incentivize the use of sustainable materials. Regulatory tools such as green certification standards, environmental impact thresholds, and minimum content requirements for recycled or bio-based materials could help institutionalize sustainable practices across the industry.

Financial mechanisms must also play a role. Subsidies, tax relief programs, and green procurement policies can lower the economic barriers that currently limit the adoption of sustainable materials. In addition, expanding access to low-interest financing for green construction projects—particularly those that incorporate bamboo, low-carbon concrete, or recycled content—would encourage developers to prioritize environmental performance without compromising economic viability.

Education and capacity building are equally essential. Training programs aimed at architects, engineers, contractors, and policymakers should emphasize life cycle thinking, material innovation, and sustainable design methodologies. These initiatives can foster a more sustainability-literate workforce capable of implementing and advocating for greener construction methods. Enhancing local production capacity is another strategic imperative. Investment in domestic manufacturing facilities for eco-materials, coupled with targeted support for small- and medium-sized enterprises (SMEs), can ensure that materials like bamboo and low-carbon concrete are available, affordable, and scalable. Encouraging public-private partnerships for research and development will be critical in tailoring material solutions to the unique climatic and socio-economic conditions of Nigeria.

Overall, this study demonstrates that sustainable building materials—when selected based on localized LCA results and integrated into broader policy, economic, and educational frameworks—can significantly reduce the environmental footprint of Nigeria’s construction industry. The findings confirm the viability of bamboo as an optimal low-impact material, support the adoption of low-carbon concrete as a transitional strategy, and validate recycled steel's role in advancing circular construction models. The strategic recommendations offered herein provide a roadmap for policymakers and industry stakeholders seeking to align construction practices with Nigeria’s sustainable development goals.

5. CONCLUSION

This study employed a life cycle assessment (LCA) approach to evaluate the environmental impacts of three sustainable building materials—bamboo, recycled steel, and low-carbon concrete—within the Nigerian construction industry. The analysis revealed that bamboo consistently outperforms the other materials across all impact categories, including global warming potential, resource depletion, acidification, and eutrophication. Its rapid renewability, low embodied energy, and minimal emissions profile highlight its promise as a cornerstone of sustainable construction in Nigeria. Even when subjected to sensitivity and scenario modeling, bamboo’s environmental superiority remained evident.

Low-carbon concrete emerged as a valuable intermediary solution. Its partial substitution of Portland cement with supplementary cementitious materials reduced both its carbon footprint and resource intensity. Although not as environmentally efficient as bamboo, its structural versatility and durability justify its inclusion in green construction practices. Recycled steel, while supportive of circular economy principles, was associated with the highest environmental burdens due to its energy-intensive recycling process and long transport distances. The study's findings underscore a critical opportunity for Nigeria to reduce construction-sector emissions and resource consumption by embracing sustainable materials. However, the path to adoption is hindered by systemic challenges including high initial costs, limited market availability, and inadequate production infrastructure. Strategic interventions are essential to overcome these barriers

Policy frameworks must be strengthened to embed sustainability into national construction codes, supported by incentives such as tax credits, low-interest loans, and subsidies for green projects. Regulatory mandates can accelerate the market transition by setting minimum environmental performance benchmarks for building materials. Capacity building through professional training and public education campaigns will also be pivotal in shifting industry mindsets and practices. Enhancing local production capabilities is a strategic imperative. Investing in domestic manufacturing, particularly for bamboo-based composites and alternative binders for low-carbon concrete, will reduce costs, enhance supply reliability, and stimulate local economies. Partnerships between government, academia, and the private sector will be key in driving innovation and developing scalable technologies tailored to regional needs.

Future research should expand the LCA database with region-specific inputs to improve the accuracy and applicability of sustainability assessments. Longitudinal studies assessing the in-use performance, durability, and cost-effectiveness of sustainable materials under Nigerian climatic and operational conditions are also recommended. Moreover, integrated assessments that combine environmental, economic, and social dimensions will offer a more holistic understanding of sustainability trade-offs in the built environment.

REFERENCES

[1] A. Akinyemi and O. Ogundare, “Environmental Sustainability Challenges in Nigeria’s Construction Sector: Trends and Impacts,” Journal of Building Performance, vol. 13, no. 2, pp. 85–95, 2022.

[2] P. Tetteh and R. King, “Green Innovation and Material Selection for Sustainable Housing in Sub-Saharan Africa,” *International Journal of Sustainable Built Environment*, vol. 10, no. 1, pp. 21–35, 2021, DOI: 10.1016/j.ijsbe.2021.03.004.

[3] Y. Geng, J. Dong, B. Fujita, and M. Fujita, “Sustainable Material Transitions in the Construction Sector: A Circular Economy Perspective,” *Journal of Cleaner Production*, vol. 270, pp. 122-142, 2020, DOI: 10.1016/j.jclepro.2020.122142.

[4] United Nations Environment Programme (UNEP), *2022 Global Status Report for Buildings and Construction*, Nairobi: UNEP, 2022.

[5] World Green Building Council, *Bringing Embodied Carbon Upfront: Coordinated Action for the Building and Construction Sector to Tackle Embodied Carbon*, 2020.

[6] A. Ogunde, T. Olaniran, and I. Adegbite, “Barriers to the Adoption of Sustainable Construction Practices in Nigeria,” *Journal of Sustainable Development in Africa*, vol. 25, no. 3, pp. 44–58, 2023.

[7] O. Olugbenga and A. Akinlolu, "Policy Gaps in the Promotion of Green Building in Nigeria," *Environmental Management and Sustainable Development*, vol. 10, no. 1, pp. 67–83, 2021, DOI: 10.5296/emsd.v10i1.18263.

[8] International Organization for Standardization (ISO), "ISO 14040: Environmental Management — Life Cycle Assessment — Principles and Framework," Geneva: ISO, 2020.

[9] T. Adewumi, K. Sanni, and F. Ajayi, "Assessing Local Approaches to Sustainable Building in Nigeria: Opportunities and Challenges," *Journal of Environmental Engineering and Management*, vol. 17, no. 4, pp. 92–106, 2023.

[10] A. Oke, S. Adeniran, and K. Lawal, "Life Cycle Thinking and Green Construction Practices in Nigeria," *Built Environment Journal*, vol. 29, no. 2, pp. 34–49, 2022.

[11] L. Li, B. Huang, and C. Cheng, "Mechanical and Environmental Assessment of Bamboo as a Sustainable Building Material," *Journal of Cleaner Production*, 2021, vol. 278, DOI: 10.1016/j.jclepro.2020.123984.

[12] T. Adewumi, K. Sanni, and F. Ajayi, "Assessing Local Approaches to Sustainable Building in Nigeria: Opportunities and Challenges," *Journal of Environmental Engineering and Management*, 2023, vol. 17, no. 4, pp. 92–106.

[13] H. Fufa, M. Zeb, and Y. Baek, "Circular Economy in Steel Manufacturing: LCA-Based Analysis of Recycled Steel," *Sustainability*, 2023, vol. 15, no. 2, DOI: 10.3390/su15020982.

[14] Y. Zhang and H. Xiao, "Life Cycle Emissions and Environmental Payback of Recycled Steel in Construction," *Journal of Construction and Building Materials*, 2023, vol. 360, DOI: 10.1016/j.conbuildmat.2022.129696.

[15] Y. Liu, F. Chen, and Z. Xue, "Performance of Recycled Aggregate in Concrete: A Review," *Resources, Conservation and Recycling*, 2023, vol. 190, DOI: 10.1016/j.resconrec.2022.106754.

[16] K. Scrivener, V. John, and E. Gartner, "Eco-efficient Cements: Potential, Economical and Environmental Benefits," *Cement and Concrete Research*, 2018, vol. 114, pp. 2–26, DOI: 10.1016/j.cemconres.2018.03.015.

[17] S. Ahmad, M. Irfan, and N. Z. Khan, "A Review of Low-Carbon Concrete: Pathways for Sustainable Infrastructure," *Construction and Building Materials*, 2023, vol. 345, DOI: 10.1016/j.conbuildmat.2022.128478.

[18] P. Tetteh and R. King, "Green Innovation and Material Selection for Sustainable Housing in Sub-Saharan Africa," *International Journal of Sustainable Built Environment*, 2021, vol. 10, no. 1, pp. 21–35, DOI: 10.1016/j.ijsbe.2021.03.004.

[19] J. Ortiz-Rodriguez, M. Diaz-Lopez, and C. Hernandez, "Bio-Based Insulation Materials in Green Construction: Life Cycle Perspective," *Energy and Buildings*, 2022, vol. 263, DOI: 10.1016/j.enbuild.2022.112042.

[20] I. Khatib, "Lifecycle Cost and Environmental Impact of Green Building Materials," *Environmental Engineering and Management Journal*, 2020, vol. 19, no. 12, pp. 2167–2175, DOI: 10.30638/eemj.2020.204.

[21] United Nations Environment Programme (UNEP), *2022 Global Status Report for Buildings and Construction*, Nairobi: UNEP, 2022.

[22] ISO, "ISO 14040: Environmental Management – Life Cycle Assessment – Principles and Framework," International Organization for Standardization, Geneva, 2020.

[23] M. A. Curran, *Life Cycle Assessment Handbook: A Guide for Environmentally Sustainable Products*, Hoboken, NJ: John Wiley & Sons, 2012.

[24] U.S. Green Building Council (USGBC), "LEED v4 for Building Design and Construction," Washington, D.C.: USGBC, 2020.

[25] O. Ogunde, T. Olaniran, and I. Adegbite, "Barriers to the Adoption of Sustainable Construction Practices in Nigeria," *Journal of Sustainable Development in Africa*, 2023, vol. 25, no. 3, pp. 44–58.

[26] M. Shad, L. Wang, and H. Gao, "Comparative Life Cycle Assessment of Building Materials: Trends and Challenges," *Environmental Impact Assessment Review*, 2023, vol. 98, DOI: 10.1016/j.eiar.2023.106974.

[27] A. Oke, S. Adeniran, and K. Lawal, "Life Cycle Thinking and Green Construction Practices in Nigeria," *Built Environment Journal*, 2022, vol. 29, no. 2, pp. 34–49.

[28] Nigerian Building and Road Research Institute (NBRRI), *Research and Development in Sustainable Building Materials and Construction Technologies*, Abuja: NBRRI Press, 2022.

[29] A. Dodoo, L. Gustavsson, and R. Sathre, "Life cycle primary energy use and carbon emission of conventional and low-energy buildings: A Swedish case study," *Energy and Buildings*, 2014, vol. 81, pp. 282–294, DOI: 10.1016/j.enbuild.2014.06.035.

[30] M. M. Bilec, M. Guggemos, and M. R. Ries, "Life-cycle assessment tool for construction: A case study of sustainability metrics for concrete," *Journal of Infrastructure Systems*, 2010, vol. 16, no. 4, pp. 266–275, DOI: 10.1061/(ASCE)IS.1943-555X.0000036.

[31] P. Penna, M. Guerrini, and M. Bragadin, "Environmental and economic assessment of retrofit strategies for buildings," *Energy and Buildings*, 2015, vol. 95, pp. 57–65, DOI: 10.1016/j.enbuild.2015.02.021.

[32] D. M. Vieira and A. Horvath, "Comparison of environmental impacts of aggregate production using recycled and natural aggregates," *Resources, Conservation and Recycling*, 2008, vol. 52, no. 2, pp. 197–204, DOI: 10.1016/j.resconrec.2007.06.002.

[33] S. Hellweg and L. Milà i Canals, "Emerging approaches, challenges and opportunities in life cycle assessment," *Science*, 2014, vol. 344, no. 6188, pp. 1109–1113, DOI: 10.1126/science.1248361.

[34] A. Adebayo, "Sustainable housing delivery in Nigeria: A framework for institutional roles," *Journal of*

Environmental Design and Management, 2014, vol. 6, no. 2, pp. 35–48.

[35] A. Akinola and R. Okunola, "The challenges of sustainable building practices in Nigeria," *Nigerian Journal of Sustainable Development*, 2014, vol. 9, no. 1, pp. 21–29.

[36] J. Fapohunda and P. Stephenson, "Optimal management of the construction workforce towards sustainable construction," *International Journal of Project Management*, 2010, vol. 28, no. 8, pp. 841–849, DOI: 10.1016/j.ijproman.2010.01.003.

[37] H. Ahmed and A. Solomon, "Mixed methods in construction sustainability: Integrating empirical and contextual insights," *Built Environment Research Journal*, 2024, vol. 18, no. 1, pp. 44–59.

[38] L. Ciaccioni, M. Marino, and F. Torresi, "Triangulation in building performance research: A methodological review," *Journal of Sustainable Architecture*, 2025, vol.13, no. 2, pp. 22–39.

[39] F. Reichert, S. Meyer, and D. Hufnagel, "Sustainable building performance: Combining LCA with field-based evidence," *Energy and Buildings*, 2025, vol. 258, DOI: 10.1016/j.enbuild.2022.111831.

[40] E. Akpan, "Holistic Assessment of Building Materials in Nigeria: A Mixed Methods Approach," *African Journal of Construction Science*, 2025, vol. 11, no. 1, pp. 14–27.

[41] H. C. O. Unegbu and D. S. Yawas, "Optimizing construction and demolition waste management in Nigeria: Challenges, regulatory frameworks, and policy solutions," *Discover Civil Engineering*, vol. 1, no. 1, p. 141, 2024.

[42] H. Unegbu and D. S. Yawas, "Assessing the impact of green building certifications on construction practices in Nigeria: A systematic review," *Indonesian Journal of Engineering and Technology (INAJET)*, vol. 7, no. 1, pp. 7–14, 2024.

[43] H. C. O. Unegbu, D. S. Yawas, B. Dan-Asabe, and A. A. Alabi, "Innovative energy-efficient solutions for sustainable development in Nigeria's construction industry," *MECHANICAL*, vol. 15, no. 2, p. 199, 2024.

[44] H. C. O. Unegbu, D. S. Yawas, B. Dan-Asabe, and A. A. Alabi, "Measures for overcoming sustainable construction barriers in the Nigerian construction industry," *Discover Civil Engineering*, vol. 2, no. 1, p. 26, 2025.

[45] T. Chotisarn and R. Phuthong, "Integrating qualitative data in LCA-based material research: Lessons from Southeast Asia," *Sustainability Research Letters*, 2025, vol. 11, no. 2, pp. 88–104.

[46] O. Olugbenga and A. Akinlolu, "Policy Gaps in the Promotion of Green Building in Nigeria," *Environmental Management and Sustainable Development*, 2019, vol. 10, no. 1, pp. 67–83, DOI: 10.5296/emsd.v10i1.18263.

[47] L. F. Cabeza, A. Castell, C. Medrano, I. Martorell, G. Pérez, and M. Urigüen, "Life Cycle Assessment (LCA) and Life Cycle Energy Analysis (LCEA) of Buildings and the Building Sector: A Review," *Renewable and Sustainable Energy Reviews*, 2014, vol. 29, pp. 394–416, DOI: 10.1016/j.rser.2013.08.037.

[48] M. Asif, T. Muneer, and R. Kelley, "Life Cycle Assessment: A Case Study of a Dwelling Home in Scotland," *Building and Environment*, 2007, vol. 42, no. 3, pp. 1391–1394, DOI: 10.1016/j.buildenv.2005.11.023.

[49] C. J. Kibert, *Sustainable Construction: Green Building Design and Delivery*, 3rd ed., Hoboken, NJ: John Wiley & Sons, 2016.

[50] ISO, "ISO 14040: Environmental Management – Life Cycle Assessment – Principles and Framework," International Organization for Standardization, Geneva, 2006.

[51] J. B. Guinée, *Handbook on Life Cycle Assessment: Operational Guide to the ISO Standards*, Dordrecht: Springer, 2011.

[52] W. Khasreen, M. Banfill, and G. Menzies, "Life-Cycle Assessment and the Environmental Impact of Buildings: A Review," *Sustainability*, 2009, vol. 1, no. 3, pp. 674–701, DOI: 10.3390/su1030674.

[53] J. Ortiz, F. Castells, and G. Sonnemann, "Sustainability of Construction Materials: LCA-based Performance Analysis," *Building and Environment*, 2009, vol. 44, no. 3, pp. 393–401, DOI: 10.1016/j.buildenv.2008.03.006.

[46] O. Olugbenga and A. Akinlolu, "Policy Gaps in the Promotion of Green Building in Nigeria," *Environmental Management and Sustainable Development*, 2019, vol. 10, no. 1, pp. 67–83, DOI: 10.5296/emsd.v10i1.18263.

[47] L. F. Cabeza, A. Castell, C. Medrano, I. Martorell, G. Pérez, and M. Urigüen, "Life Cycle Assessment (LCA) and Life Cycle Energy Analysis (LCEA) of Buildings and the Building Sector: A Review," *Renewable and Sustainable Energy Reviews*, 2014, vol. 29, pp. 394–416, DOI: 10.1016/j.rser.2013.08.037.

[48] M. Asif, T. Muneer, and R. Kelley, "Life Cycle Assessment: A Case Study of a Dwelling Home in Scotland," *Building and Environment*, 2007, vol. 42, no. 3, pp. 1391–1394, DOI: 10.1016/j.buildenv.2005.11.023.

[49] C. J. Kibert, *Sustainable Construction: Green Building Design and Delivery*, 3rd ed., Hoboken, NJ: John Wiley & Sons, 2016.

[50] ISO, "ISO 14040: Environmental Management – Life Cycle Assessment – Principles and Framework," International Organization for Standardization, Geneva, 2006.

[51] J. B. Guinée, *Handbook on Life Cycle Assessment: Operational Guide to the ISO Standards*, Dordrecht: Springer, 2011.

[52] W. Khasreen, M. Banfill, and G. Menzies, "Life-Cycle Assessment and the Environmental Impact of Buildings: A Review," *Sustainability*, 2009, vol. 1, no. 3, pp. 674–701, DOI: 10.3390/su1030674.

[53] J. Ortiz, F. Castells, and G. Sonnemann, "Sustainability of Construction Materials: LCA-based Performance

Analysis," *Building and Environment*, 2009, vol. 44, no. 3, pp. 393–401, DOI: 10.1016/j.buildenv.2008.03.006.
[54] ISO, "ISO 14044: Environmental Management – Life Cycle Assessment – Requirements and Guidelines," International Organization for Standardization, Geneva, 2006.
[55] F. J. Fowler, *Survey Research Methods*, 6th ed., Thousand Oaks, CA: Sage Publications, 2020.
[56] A. Fufa, A. Bekele, and B. Chewaka, "Comparative Life Cycle Inventory of Building Materials for Low Carbon Construction," *Journal of Environmental Management*, vol. 341, 2023, 118184, DOI: 10.1016/j.jenvman.2023.118184.
[57] A. Ahmad, M. K. Arif, and S. Abbas, "Contextualizing LCA Models for Sub-Saharan Africa: A Nigerian Construction Case Study," *Sustainable Infrastructure*, vol. 9, no. 1, 2023, pp. 22–34.
[58] M. Huijbregts, Z. Steinmann, and P. van Zelm, "ReCiPe 2016: A Harmonised Life Cycle Impact Assessment Method at Midpoint and Endpoint Level," *International Journal of Life Cycle Assessment*, vol. 22, no. 2, 2017, pp. 138–147, DOI: 10.1007/s11367-016-1246-y.
[59] M. Shad, L. Wang, and H. Gao, "Hotspot Identification in LCA of Emerging Building Materials," *Environmental Impact Assessment Review*, vol. 98, 2023, 106974, DOI: 10.1016/j.eiar.2023.106974.
[60] Y. Chen, X. Li, and H. Song, "Comparative Analysis of Bamboo and Timber in Sustainable Construction," *Journal of Cleaner Production*, vol. 310, 2021, 127417, DOI: 10.1016/j.jclepro.2021.127417.
[61] L. Yu, X. Wang, and H. Zhang, "Carbon Sequestration Potential of Bamboo in Green Buildings," *Sustainable Materials and Technologies*, vol. 32, 2024, e00547, DOI: 10.1016/j.susmat.2023.e00547.
[62] A. Rahman, T. Islam, and M. Kamal, "Life Cycle Assessment of Low-Carbon Concrete Using Fly Ash and Slag in Tropical Regions," *Construction and Building Materials*, vol. 351, 2023, 129303, DOI: 10.1016/j.conbuildmat.2022.129303.
[63] C. Zhao and B. Li, "Assessment of Cementitious Materials with Supplementary Components for Carbon Reduction," *Journal of Building Engineering*, vol. 57, 2022, 104886, DOI: 10.1016/j.jobe.2022.104886.
[64] P. Reddy, A. Singh, and K. Bansal, "Environmental Implications of Recycled Steel in Construction Applications," *Resources, Conservation and Recycling*, vol. 180, 2022, 106175, DOI: 10.1016/j.resconrec.2022.106175.
[65] J. Wang, M. Zhou, and L. Lin, "Circular Economy Metrics in Steel Construction: Life Cycle Trade-Offs and Policy Implications," *Journal of Industrial Ecology*, vol. 27, no. 1, 2023, pp. 120–136, DOI: 10.1111/jiec.13218.